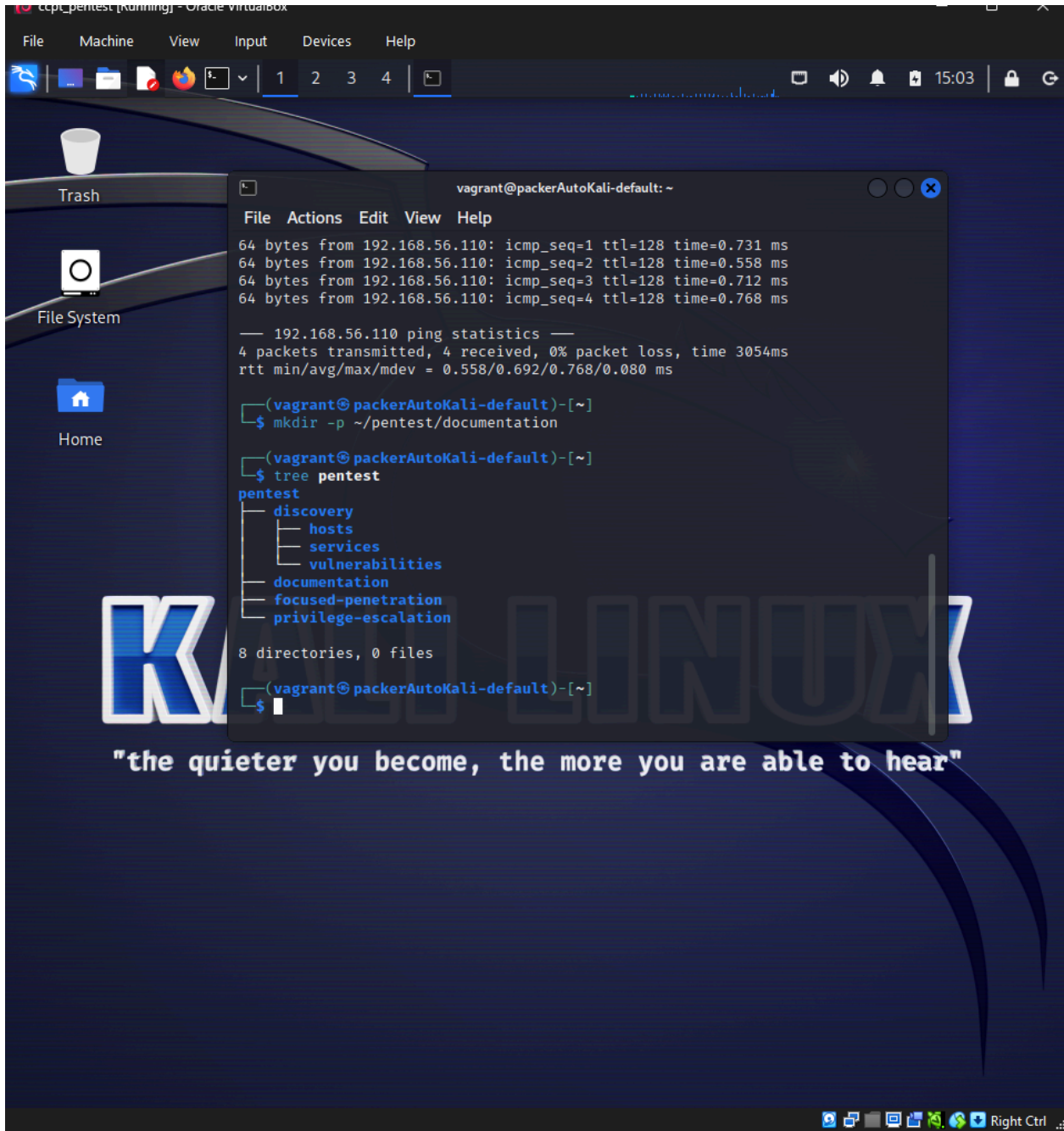


Exercise 2.1

B1: Tạo các file theo mẫu



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the results of a ping command to 192.168.56.110, showing 4 packets transmitted with 0% loss. It then shows the creation of a directory structure for a penetration test project.

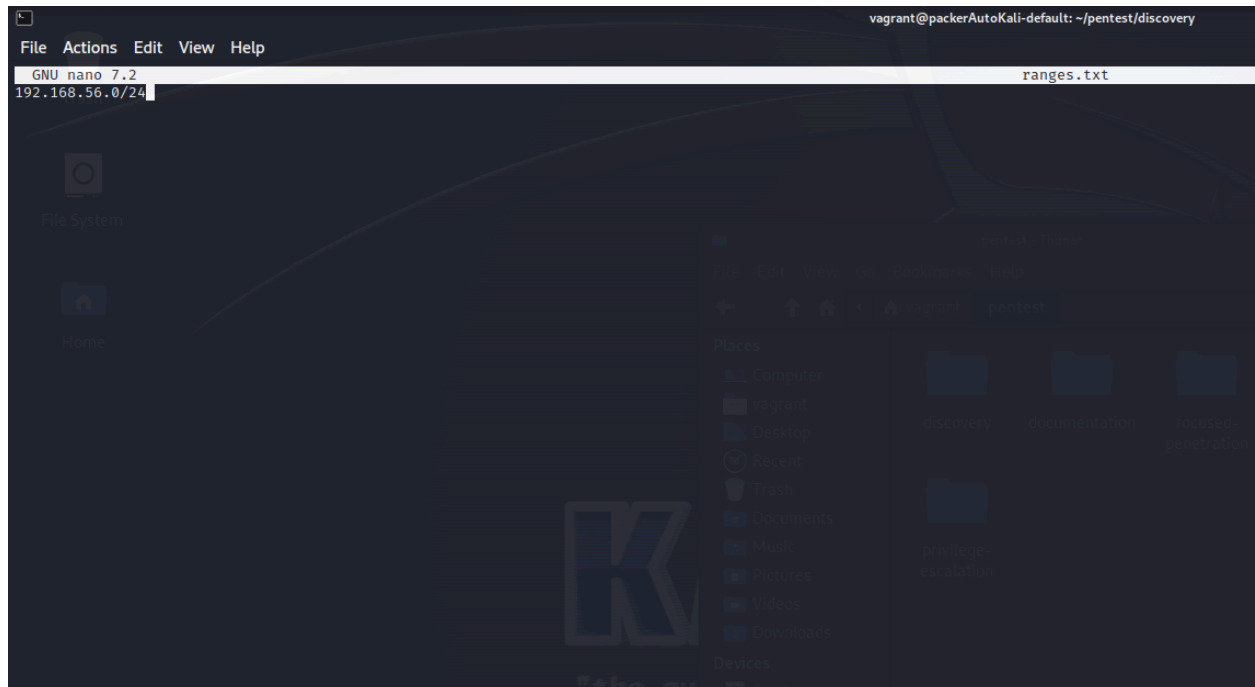
```
vagrant@packerAutoKali-default: ~  
File Actions Edit View Help  
64 bytes from 192.168.56.110: icmp_seq=1 ttl=128 time=0.731 ms  
64 bytes from 192.168.56.110: icmp_seq=2 ttl=128 time=0.558 ms  
64 bytes from 192.168.56.110: icmp_seq=3 ttl=128 time=0.712 ms  
64 bytes from 192.168.56.110: icmp_seq=4 ttl=128 time=0.768 ms  
  
— 192.168.56.110 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3054ms  
rtt min/avg/max/mdev = 0.558/0.692/0.768/0.080 ms  
  
(vagrant@packerAutoKali-default)-[~]  
$ mkdir -p ~/pentest/documentation  
  
(vagrant@packerAutoKali-default)-[~]  
$ tree pentest  
pentest  
├── discovery  
│   ├── hosts  
│   ├── services  
│   └── vulnerabilities  
├── documentation  
├── focused-penetration  
└── privilege-escalation  
  
8 directories, 0 files  
  
(vagrant@packerAutoKali-default)-[~]  
$
```

B2: Sử dụng lệnh “*ip a*” để tìm subnet, ta tìm được subnet: 192.168.56.0/24

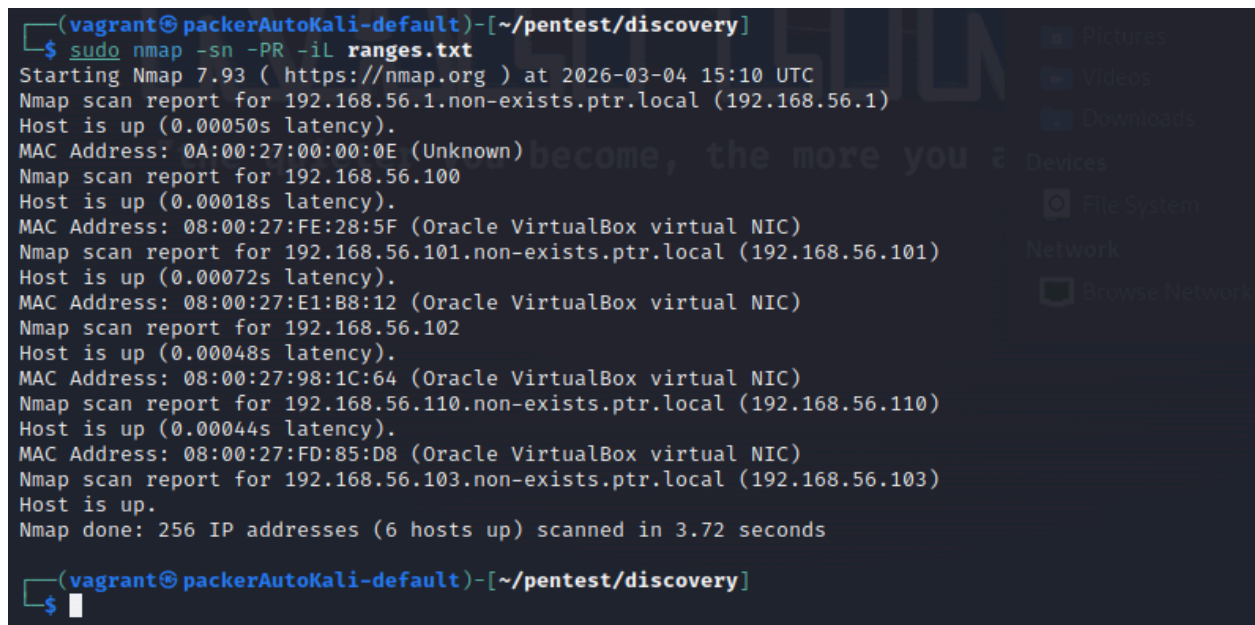
```
(vagrant@packerAutoKali-default)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    aut qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:94:69:49 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic eth0
        valid_lft 85475sec preferred_lft 85475sec
    inet6 fd17:625c:f037:2:a00:27ff:fe94:6949/64 scope global dynamic mngtmpa
    ddr
        valid_lft 86006sec preferred_lft 14006sec
    inet6 fe80::a00:27ff:fe94:6949/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:98:7e:79 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixro
    ute eth1
        valid_lft 574sec preferred_lft 574sec
    inet6 fe80::a00:27ff:fe98:7e79/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state
    DOWN group default
    link/ether 02:42:a2:0b:db:f4 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(vagrant@packerAutoKali-default)-[~]
$
```

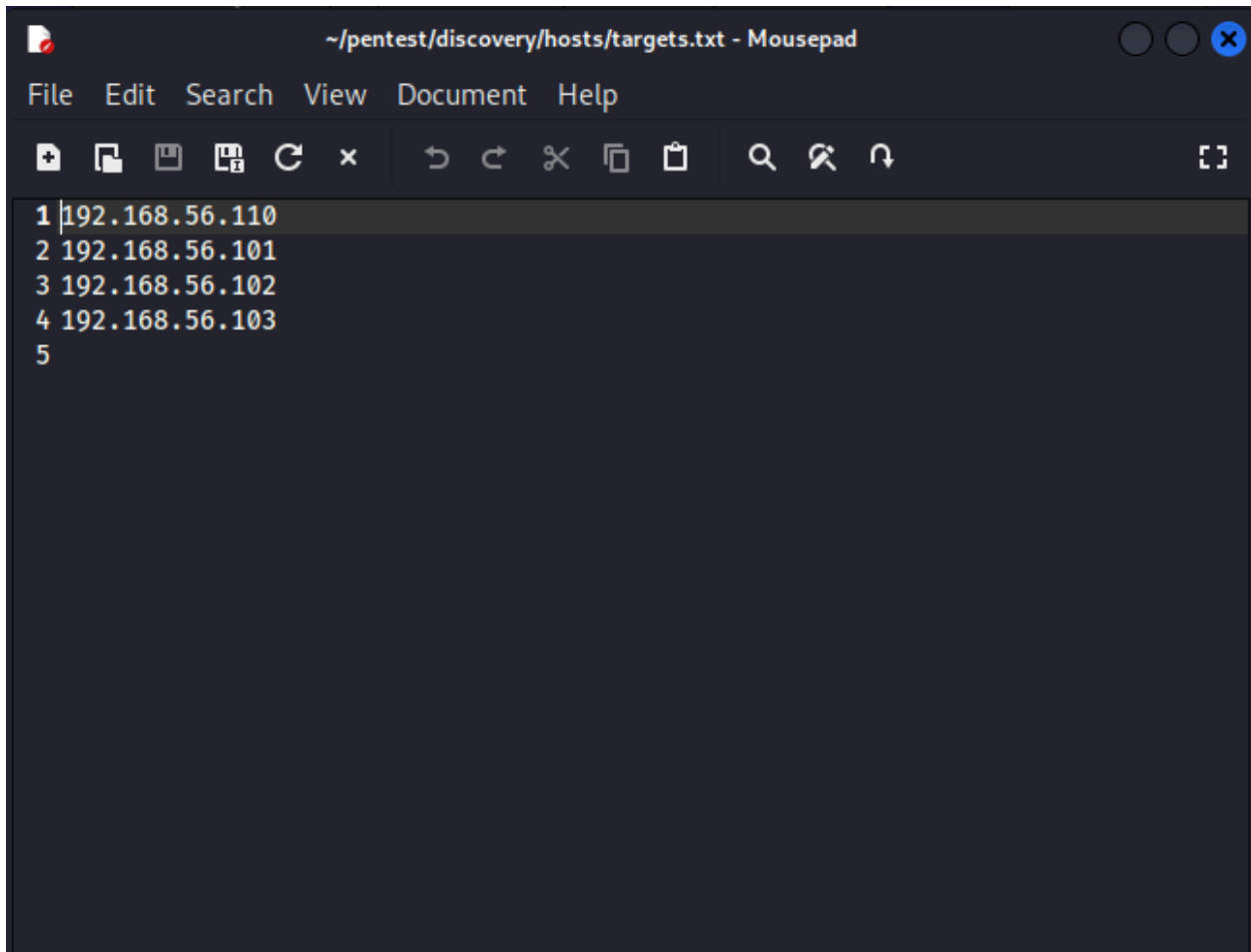
B3: Cd đến file discovery, dùng lệnh “*nano ranges.txt*” ,nhập subnet vào file và lưu



B4: Dùng lệnh “*sudo nmap -sn -PR -iL ranges.txt*” để tìm live hosts và đưa các ip vào file targets.txt



Kết quả:



The screenshot shows a text editor window titled "~/.pentest/discovery/hosts/targets.txt - Mousepad". The window has a menu bar with "File", "Edit", "Search", "View", "Document", and "Help". Below the menu bar is a toolbar with various icons for file operations and editing. The main text area contains a list of IP addresses, each preceded by a number from 1 to 5. The first line is highlighted.

```
1 192.168.56.110
2 192.168.56.101
3 192.168.56.102
4 192.168.56.103
5
```

Exercise 3.1

B1: Chạy full sweep có service detection bằng lệnh

“sudo nmap -p- -sS -sV -T4 -iL hosts/targets.txt -oA services/full-sweep”

```

Nmap scan report for 192.168.56.101
Host is up (0.00040s latency).
Not shown: 65522 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
22/tcp    open  ssh              OpenSSH for_Windows_9.5 (protocol 2.0)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
3389/tcp  open  ssl/ms-wbt-server?
5985/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
47001/tcp open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49664/tcp open  msrpc            Microsoft Windows RPC
49665/tcp open  msrpc            Microsoft Windows RPC
49666/tcp open  msrpc            Microsoft Windows RPC
49667/tcp open  msrpc            Microsoft Windows RPC
49669/tcp open  msrpc            Microsoft Windows RPC
49670/tcp open  msrpc            Microsoft Windows RPC
MAC Address: 08:00:27:E1:B8:12 (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for 192.168.56.102.non-exists.ptr.local (192.168.56.102)
Host is up (0.00037s latency).
Not shown: 65522 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
22/tcp    open  ssh              OpenSSH for_Windows_9.5 (protocol 2.0)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
3389/tcp  open  ssl/ms-wbt-server?
5985/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
47001/tcp open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49664/tcp open  msrpc            Microsoft Windows RPC
49665/tcp open  msrpc            Microsoft Windows RPC
49666/tcp open  msrpc            Microsoft Windows RPC
49667/tcp open  msrpc            Microsoft Windows RPC
49668/tcp open  msrpc            Microsoft Windows RPC
49670/tcp open  msrpc            Microsoft Windows RPC
MAC Address: 08:00:27:98:1C:64 (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for 192.168.56.103.non-exists.ptr.local (192.168.56.103)
Host is up (0.0000050s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh              OpenSSH 9.2p1 Debian 2 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 4 IP addresses (4 hosts up) scanned in 143.88 seconds

```

B2: Tạo all-ports.csv bằng

"cat services/full-sweep.gnmap | grep open > services/all-ports.csv"

Lọc HTTP servers → web.txt bằng

"grep http services/full-sweep.nmap | grep open | cut -d " " -f 2 | sort -u > hosts/web.txt"

Lọc Microsoft SQL → mssql.txt bằng

"grep -i "ms-sql" services/full-sweep.nmap | grep open | cut -d " " -f 2 | sort -u > hosts/mssql.txt"

Lọc Windows machines → windows.txt bằng

"grep -E "445/tcp|3389/tcp" services/full-sweep.nmap | grep open | cut -d " " -f 2 | sort -u > hosts/windows.txt"

Kết quả

```
(vagrant@packerAutoKali-default)-[~/pentest/discovery]
$ cat services/full-sweep.gnmap | grep open > services/all-ports.csv
File System
(vagrant@packerAutoKali-default)-[~/pentest/discovery]
$ grep http services/full-sweep.nmap | grep open | cut -d" " -f 2 | sort -u
> hosts/web.txt
cut: the delimiter must be a single character
Try 'cut --help' for more information.

(vagrant@packerAutoKali-default)-[~/pentest/discovery]
$ grep http services/full-sweep.nmap | grep open | cut -d " " -f 2 | sort -u
> hosts/web.txt

(vagrant@packerAutoKali-default)-[~/pentest/discovery]
$ grep -i "ms-sql" services/full-sweep.nmap | grep open | cut -d " " -f 2 |
sort -u > hosts/mssql.txt

(vagrant@packerAutoKali-default)-[~/pentest/discovery]
$ grep -E "455/tcp|3389/tcp" services/full-sweep.nmap | grep open | cut -d
" " -f 2 | sort -u > hosts/windows.txt
```

Kết quả tree ~/pentest

```
(vagrant@packerAutoKali-default)-[~/pentest/discovery]
$ tree ~/pentest
/home/vagrant/pentest
├── discovery
│   ├── hosts
│   │   ├── mssql.txt
│   │   ├── targets.txt
│   │   ├── web.txt
│   │   └── windows.txt
│   ├── ranges.txt
│   └── services
│       ├── all-ports.csv
│       ├── full-sweep.gnmap
│       ├── full-sweep.nmap
│       └── full-sweep.xml
├── vulnerabilities
├── documentation
├── focused-penetration
└── privilege-escalation
```